MAJID HAJI BAGHERI

Device Engineering · Energy Harvesting & Storage · MEMS & Nanofabrication · Materials R&D · Hardware–ML Integration

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Work Authorization: Canadian Open Work Permit · Available part-time now; full-time from November 2026

PROFESSIONAL SUMMARY

Device-engineering researcher (PhD, Electrical & Computer Engineering) who takes energy-harvesting and sensor concepts from physics first principles through cleanroom fabrication to working, characterized prototypes. Six years of graduate R&D, including 3+ years leading industry-collaborative projects with the National Research Council of Canada (NRC) and Shimco North America — turning partner objectives into delivered devices, milestone reports, and peer-reviewed results. First author of multiple papers in Nano Energy, Energy & Environmental Materials, and Advanced Sensor Research, and co-author in Nature Communications (16 peer-reviewed publications total). Pairs hands-on micro/nanofabrication and multi-modal characterization with DFT/COMSOL simulation and Python-based machine learning. A condensed-matter physics foundation drives strong modeling, instrumentation, and quantitative problem-solving applied directly to device performance.

AREAS OF EXPERTISE

- **Energy Harvesting & Sensors:** Piezoelectric/triboelectric nanogenerators (PENGs/TENGs); self-powered, wearable & cryogenic-compatible sensor systems; lead-free perovskite piezocomposites; lithium-ion battery anodes.
- **Microfabrication & Cleanroom:** Photolithography; sputtering; CVD/PVD/CVT; spin coating; electrospinning; thin-film deposition; etching; bonding; MEMS/diode/MOSFET fabrication; class-controlled cleanroom operation.
- **Characterization:** XRD; FTIR; SEM/EDX; TEM; AFM; optical microscopy; I–V, C–V; cyclic voltammetry; galvanostatic cycling; electromechanical & electroacoustic testing.
- **Simulation & Modeling:** Quantum ESPRESSO (DFT, electronic structure, ferroelectric polarization); COMSOL Multiphysics; Ansys Lumerical; Monte Carlo / stochastic growth (KPZ); MATLAB.
- **Programming & Data:** Python (signal processing, ML classification on sensor data); MATLAB; Origin Pro; LaTeX; Microsoft Office.
- **Leadership & Delivery:** Industry/government partner management; milestone reporting; technical presentations; mentoring of co-op & undergraduate researchers.
- **Certification:** Certified Radiation Protection Professional — Non-Destructive Testing (NDT), Medical & Industrial.

PROFESSIONAL EXPERIENCE

Graduate Researcher (PhD) — University of Waterloo · Prof. Dayan Ban's Lab · Waterloo, ON — Sept 2022 – Present

- Led three concurrent partner-driven R&D streams from concept to publication, delivering devices, milestone reports, and technical presentations to the **National Research Council of Canada (NRC)** and **Shimco North America**.
- Designed and fabricated piezoelectric and triboelectric nanogenerators (PENGs/TENGs) for self-powered sensing and energy harvesting, producing first-author results published in *Nano Energy* and *Energy & Environmental Materials*.
- Built a TENG-based acoustic sensor coupled to a machine-learning classifier for real-time sound monitoring and captioning — published in *Advanced Sensor Research* and presented at IEEE CCECE.
- Engineered a multi-layer cascaded nanogenerator architecture that operated reliably under cryogenic and low-temperature conditions, extending self-powered devices into harsh-environment use cases.
- Ran first-principles DFT simulations (Quantum ESPRESSO) on lead-free perovskites to predict ferroelectric polarization and guide material selection, linking electronic-structure modeling to measured device output.
- Executed end-to-end fabrication — photolithography, thin-film deposition, electrode patterning, encapsulation — and full electromechanical characterization.
- Designed and built custom motor-tapping and cryogenic test benches that enabled reproducible benchmarking and de-risked downstream device development.
- Mentored co-op and undergraduate research assistants; co-authored perovskite piezocomposite work in *Nature Communications* that was featured on CTV News Kitchener and University of Waterloo News.

Graduate Researcher (MSc) — Sharif University of Technology · Battery Research Group · Tehran, Iran — Feb 2021 – Feb 2022

- Led the battery research group — coordinated experimental planning, project workflows, and a student team across lithium-ion anode projects.
- Designed, built, and programmed a custom inert-atmosphere glove box for moisture-sensitive Li-ion battery fabrication, enabling reliable in-house cell assembly (sealing, atmosphere control, instrumentation).
- Synthesized and characterized SnS_2-based nanostructured anodes and freestanding $Ti_3C_2T_x$ MXene / graphene-oxide papers for energy storage.

- Improved cell performance through binder engineering (PEDOT:PSS, PVDF, CMC), targeting capacity retention and cycling stability.
- Delivered industry-collaborative R&D with **Parthian Battery Novin** and the **Iran Nano Battery Network** on self-healing MXene anode materials (Sharif–industry project, begun as undergraduate researcher 2019–2021).

EDUCATION

Ph.D., Electrical & Computer Engineering (Nanotechnology) · University of Waterloo, Canada · GPA 3.70/4.00 Expected Nov 2026
Thesis: Next-Generation Nanogenerators for Sustainable Energy Solutions in Challenging Environments · Advisor: Prof. Dayan Ban

M.Sc., Physics · Sharif University of Technology (SUT), Iran · GPA 3.82/4.00 2021
Thesis: Hybrid Nanostructures as Anode Material for Ionic Batteries

B.Sc., Physics — Condensed Matter · Iran University of Science & Technology (IUST) · GPA 3.76/4.00 · Ranked #1 in cohort 2018
Thesis: Monte Carlo simulation of ballistic-deposition thin-film growth (KPZ universality)

SELECTED PUBLICATIONS & RECOGNITION

Co-author on 16 peer-reviewed publications; full list at Google Scholar. Selected highlights:

- "Breaking dielectric dilemma via polymer-functionalized perovskite piezocomposite with large current-density output," *Nature Communications* 15, 9511 (2024). [co-author]
- "Self-healable, recyclable, mechanically robust vitrimer composite for high-performance TENGs and self-powered wireless electronics," *Nano Energy* (2024). [first author]
- "Advancements and challenges in molecular/hybrid perovskites for piezoelectric nanogenerators: a comprehensive review," *Nano Energy* (2023). [first author]
- "Sustainable live sound monitoring and classification via triboelectric nanogenerator and machine learning," *Energy & Environmental Materials* (2025). [first author]

Awards: Best Poster Prize, Waterloo Nanotechnology Conference (2024) · WIN Nanofellowship · Graduate Student Research Dissemination (GSRD) Award · Faculty of Engineering Award (3×) · Research Publication & Publication Excellence Awards (Prof. Ban's Lab).

TEACHING & KNOWLEDGE TRANSFER

- Invited guest lecturer, **Wilkes University (USA)**, Apr 2026 — "Triboelectric-nanogenerator microphones integrated with machine learning for sustainable sound monitoring."
- Graduate Teaching / Lab Assistant across 6+ ECE and Physics courses — numerical methods, characterization of nanomaterials, MEMS device fabrication & simulation, and microfabrication/thin-film labs (University of Waterloo & Sharif University of Technology).